THE CRITICAL MIXED-ARITHMETIC FOUR-POINT HRT THEOREM

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Editorial scope. Counted unit: the author's lemma `lem:winding`, with its complete original proof. This revision restores the author's continued-fraction and Denjoy–Koksma context (source0.tex lines 1028–1059) and the four cited bibliography entries. The statement (1067–1076) and proof (1078–1177) are unchanged. Every mathematical excerpt and bibliography entry below is copied exactly from the bundled source. The published item is preserved separately.

Author context: irrational returns and Denjoy–Koksma

For $\theta \notin \mathbb{Q}$, let

$$R_\theta x = x - \theta \pmod 1$$

act on $\mathbb{T}$. The rotation R_θ is ergodic [4]. If p_n/q_n are the continued-fraction convergents of θ, then

$$q_n\theta - p_n \longrightarrow 0. \tag{1}$$

Thus $R_\theta^{q_n}$ approaches the identity, and the denominators q_n provide the natural times at which an iterated cocycle can be compared with its initial value.

At these return times, accumulated oscillation is controlled by the Denjoy–Koksma inequality: for every real-valued periodic function g of bounded variation,

$$\sup_{x\in\mathbb{T}} \left| \sum_{j=0}^{q_n-1} g(x - j\theta) - q_n \int_{\mathbb{T}} g \right| \leq \operatorname{Var}(g). \tag{2}$$

For complex-valued g, we apply the estimate to its real and imaginary parts. In the winding argument below, it controls the derivative of the accumulated phase; later, it controls the oscillatory part of a periodic logarithm. Together with (1), it converts recurrence of the base rotation into rigidity of the cocycle. See [2, 3, 1].

1. The counted unit and the author's complete proof

Lemma 1.1 (Measurable winding obstruction)**.** *Let $\theta \notin \mathbb{Q}$, let $0 \neq h \in L^2(\mathbb{T})$, and let $b \in C^2(\mathbb{T}; \mathbb{C}^\times)$. If*

$$h(x - \theta) = b(x)h(x) \tag{3}$$

almost everywhere, then $\operatorname{wind}(b) = 0$.

Proof. **Step 1: Remove the zero set of the transfer function.** Because b never vanishes, the zero set of h is invariant, modulo null sets, under the irrational rotation. Ergodicity implies that this zero set has measure zero or one. Since $h \neq 0$, it has measure zero. Therefore

$$u(x) = \frac{h(x)}{|h(x)|}$$

is defined almost everywhere and has modulus one.

Step 2: Separate the winding from the periodic phase. Let $\psi = b/|b|$. Then

$$u(x - \theta) = \psi(x)u(x)$$

almost everywhere and $\operatorname{wind}(\psi) = \operatorname{wind}(b)$. If this common winding is $d \in \mathbb{Z}$, there is a real-valued periodic C^2 function g such that

$$\psi(x) = e^{2\pi i(dx+g(x))}.$$

Step 3: Iterate through continued-fraction returns. Iteration through q_n steps gives

$$u(x - q_n\theta) = e^{2\pi i\Phi_n(x)}u(x),$$

where

$$\Phi_n(x) = dq_n x - d\theta\frac{q_n(q_n-1)}{2} + \sum_{j=0}^{q_n-1} g(x - j\theta). \tag{4}$$

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Since $|u| = 1$,

$$|e^{2\pi i\Phi_n(x)} - 1| = |u(x - q_n\theta) - u(x)|.$$

By (1) and translation continuity in $L^2(\mathbb{T})$,

$$e^{2\pi i\Phi_n} \longrightarrow 1 \quad \text{in } L^2(\mathbb{T}). \tag{5}$$

In particular,

$$\int_{\mathbb{T}} e^{2\pi i\Phi_n(x)}\,\mathrm{d}x \longrightarrow 1. \tag{6}$$

Step 4: Exclude nonzero winding by oscillation. Assume $d \neq 0$. Differentiating (4) gives

$$\Phi_n'(x) = dq_n + \sum_{j=0}^{q_n-1} g'(x - j\theta).$$

The function g' has mean zero and bounded variation. Denjoy–Koksma therefore yields

$$\sup_x \left| \sum_{j=0}^{q_n-1} g'(x - j\theta) \right| \leq \mathrm{Var}(g').$$

Hence $\inf_x |\Phi_n'(x)| \geq cq_n$ for all large n, with $c > 0$. Moreover,

$$\|\Phi_n''\|_{L^1} \leq q_n \|g''\|_{L^1}.$$

Although Φ_n need not be periodic, it satisfies

$$\Phi_n(x + 1) = \Phi_n(x) + dq_n,$$

and Φ_n' is periodic. Consequently, the boundary term in integration by parts vanishes:

$$\left[\frac{e^{2\pi i\Phi_n(x)}}{2\pi i\Phi_n'(x)} \right]_0^1 = 0.$$

It follows that

$$\left| \int_{\mathbb{T}} e^{2\pi i\Phi_n(x)}\,\mathrm{d}x \right| \leq \frac{\|\Phi_n''\|_{L^1}}{2\pi(\inf|\Phi_n'|)^2} = O(q_n^{-1}).$$

This contradicts (6). Therefore $d = 0$. □

References

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